FINAL EXPLANATION: BIOLOGY GRADE 10

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 10 — SUB-STRAND 3.2: ANIMAL TRANSPORT	
Student Name	____
Class	____
Date	____

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INSTRUCTIONS FOR STUDENTS

This is your Final Explanation assessment. You have spent 12 lessons investigating how animal transport systems are built, how they work, and why they differ across the animal kingdom. Now it is time to bring everything together into one well-reasoned, evidence-based explanation.

Your task: Write a detailed response to the driving question —

"How does the structure of an animal's transport system determine its ability to survive and thrive — and what would happen if Eliud Kipchoge had the circulatory system of a grasshopper?"

READ CAREFULLY BEFORE YOU BEGIN:

- Answer every section fully. Each section builds on the last — together they form your complete explanation.
- Use scientific vocabulary you have learned (e.g., open/closed circulation, double circulation, haemolymph, ventricles, atria, tracheal system, gill circulation, pulmonary circuit, systemic circuit, cardiac output).
- Connect your answers back to the anchoring phenomenon — Kipchoge's heart rate, his rapid recovery, and the animals we compared him to (grasshopper, tilapia, frog).
- Support your claims with evidence from lessons, observations, diagrams, and data you collected.
- Write in full sentences and organise your ideas clearly.

Section 1: Why Do Animals Need a Transport System?

Explain why animals need a transport system. In your answer: **– Describe what substances need to be transported and why.** **– Explain how body size and complexity affect the need**	All living cells need a constant supply of oxygen and nutrients to carry out respiration and other metabolic processes, and they must also get rid of waste products like carbon dioxide and urea. In a single-celled organism, these substances can move in and out by simple diffusion across the cell membrane. However, in a large, complex animal like Eliud Kipchoge, diffusion alone is far too slow — the distance from his lungs to his leg muscles is far greater than diffusion can cover quickly enough to be useful. This is why a transport system is essential: it moves substances rapidly over long distances inside the body. During Kipchoge's marathon, his leg muscles are respiring at an enormous rate, consuming oxygen and glucose and producing carbon dioxide.

for a transport system. – **Use at least TWO specific examples from the animals studied (grasshopper, tilapia, frog, or Kipchoge as a mammal), and refer back to the anchoring phenomenon to explain what would go wrong if transport failed during a marathon.**	Without his circulatory system pumping blood continuously, those muscles would run out of oxygen within seconds, lactic acid would accumulate, and he would collapse — certainly not cross the finish line at world-record pace. Not all animals face the same challenge. A grasshopper on the school fence is much smaller and has a relatively low metabolic demand compared to a running human. Its cells are also closer to the body surface. Because of this, oxygen reaches the grasshopper's tissues directly through a network of tiny tubes called tracheae, which open at spiracles on the body surface and branch throughout the body — no blood transport of oxygen is needed at all. The grasshopper's open circulatory system, where haemolymph washes loosely around organs, is sufficient to transport nutrients and hormones but does not need to deliver oxygen at high speed. In contrast, a tilapia in Lake Victoria must move oxygen absorbed at the gills to all body tissues efficiently, requiring a closed circulatory system with a two-chambered heart. The greater the complexity and activity level of an animal, the more sophisticated its transport system must be.

Section 2: Comparing Transport Systems Across Animals	
Compare and contrast the transport systems of a grasshopper, a tilapia, a frog, and a mammal (use Kipchoge as your mammal). Your answer should address: – **The type of circulatory system (open or closed) and what that means.** – **The structure of the heart (number of chambers) in each animal.** – **How oxygen is delivered to tissues in each animal.** – **Why each system is well-matched to that animal's lifestyle and environment.** **Use a logical structure — you may compare animal by animal OR feature by feature.**	The four animals we studied show a clear pattern: as animals become more complex and active, their transport systems become more efficient and specialised. The grasshopper has an OPEN circulatory system. Its heart is a simple tube-like structure running along its back, which pumps haemolymph — a fluid that is not separated into arteries and veins — loosely around the body cavity (haemocoel). Crucially, haemolymph does NOT carry oxygen to cells. Instead, oxygen travels directly to cells through the tracheal system: a network of air tubes (tracheae) that branch from openings called spiracles on the grasshopper's exoskeleton. This works well for a small, relatively slow insect but would be completely inadequate for a large, highly active mammal. The tilapia, a fish found in Lake Victoria, has a CLOSED circulatory system — blood is always contained within vessels. Its heart has TWO chambers: one atrium and one ventricle. Blood follows a single circulatory loop: heart → gills (where oxygen is absorbed from water) → body tissues → heart. This is called single circulation. Because blood passes through the gills before reaching the body, its pressure drops, limiting how fast it reaches the muscles. This is acceptable for a fish swimming steadily in water but would not support the intense oxygen demands of a marathon runner. The frog, living in Rift Valley wetlands, has a THREE-chambered heart: two atria and one ventricle. It has a double circulation — blood travels to the lungs/skin and back (pulmonary/cutaneous circuit), and separately to the body (systemic circuit). However, because there is only one ventricle, oxygenated and deoxygenated blood mix slightly inside the heart. Frogs are ectotherms (cold-blooded) with relatively low metabolic rates, so this partial mixing is tolerable. The frog's system is more efficient than the fish's but less efficient than a mammal's. Eliud Kipchoge, as a mammal, has a FOUR-chambered heart with two atria and two ventricles, creating a complete double circulation with full separation of oxygenated and deoxygenated blood. The right side of the heart pumps deoxygenated blood to the lungs (pulmonary circuit); the left side pumps freshly oxygenated blood to the entire body (systemic circuit) at high pressure. This means Kipchoge's muscles receive blood that is rich in oxygen and at sufficient pressure to sustain rapid, high-intensity activity — enabling him to run 42 km at sub-3-minute-per-kilometre pace.

Section 3: The Mammalian Heart — Structure and Function	
Explain how the structure of the mammalian heart makes it so efficient. In your answer: – **Describe the four chambers and explain the role of each.** – **Explain the role of valves and why they matter.** – **Define cardiac output and explain how Kipchoge's heart achieves high cardiac output.** – **Explain why his heart rate recovers so quickly after a race, and what this tells us about the relationship between fitness and heart structure/function.**	The mammalian heart is a double pump divided into four chambers: the right atrium, right ventricle, left atrium, and left ventricle. The wall dividing the left side from the right side is called the septum, and it ensures that oxygenated and deoxygenated blood NEVER mix — this is the key advantage over the frog's three-chambered heart. Deoxygenated blood returning from the body enters the RIGHT ATRIUM through the vena cava. It is pushed into the RIGHT VENTRICLE, which contracts to send it to the lungs via the pulmonary artery. In the lungs, blood picks up oxygen and loses carbon dioxide. This oxygenated blood returns to the LEFT ATRIUM via the pulmonary vein, passes into the LEFT VENTRICLE — which has the thickest, most muscular wall of all four chambers — and is then pumped at high pressure to the rest of the body through the aorta. Valves (the atrioventricular valves between atria and ventricles, and the semilunar valves at the exits of the ventricles) ensure that blood flows in ONE direction only. They prevent backflow, making every heartbeat maximally efficient. Cardiac output is the volume of blood pumped by the heart per minute. It is calculated as: Cardiac Output = Heart Rate × Stroke Volume. Kipchoge's heart has adapted through years of training to have an exceptionally large stroke volume — it pumps a greater volume of blood with each beat than an untrained person's heart. This means his heart does not need to beat as rapidly to deliver the same amount of oxygen. His resting heart rate of approximately 37 beats per minute (compared to a typical 70–80 bpm) reflects this: each beat is so powerful and efficient that fewer beats are needed. After a race, Kipchoge's heart rate drops rapidly because his highly trained cardiac muscle and autonomic nervous system respond efficiently — his large stroke volume means the heart quickly meets the body's returning oxygen demands without needing to stay at a high rate. A less fit person's heart, with a smaller stroke volume, must keep beating fast for much longer to clear the oxygen debt. This is exactly what the heart-rate recovery data from the anchoring phenomenon showed us.

Section 4: What If Kipchoge Had a Grasshopper's Circulatory System?	
This is the heart of the driving question. Using everything you have learned, explain specifically and scientifically what would happen if Eliud Kipchoge attempted to run a marathon with the circulatory system of a grasshopper. Your answer must: – **Identify at least THREE specific structural differences between the two systems.** – **For each difference, explain the precise consequence for**	If Eliud Kipchoge attempted to run the Nairobi Standard Chartered Marathon with the circulatory system of a grasshopper, the consequences would be immediately catastrophic. Here is why, difference by difference: Difference 1 — Open vs. Closed Circulation: A grasshopper's haemolymph simply washes around the body cavity rather than being directed under pressure through vessels. Kipchoge's body, at 1.67 m tall and approximately 57 kg, is vastly larger than a grasshopper. Blood (or haemolymph) needs to travel metres to reach his feet and return. An open system generates almost no pressure, so fluid movement would rely mostly on body movements. During a marathon, his leg muscles demand enormous volumes of oxygen every second. With no vessels directing flow to where it is needed most, his muscles would be starved of oxygen almost immediately. He would collapse within the first few hundred metres — likely before leaving Nairobi's city centre. Difference 2 — Tracheal Oxygen Delivery vs. Blood-Based Oxygen Delivery: In a grasshopper, oxygen reaches cells directly through tracheae — this works because the grasshopper's body is small enough (a few centimetres) that tracheae can reach every cell. Kipchoge's body is far too large for tracheae to serve every cell; the diffusion distance would be impossibly long. Even if we imagined scaling up a tracheal system to human size, the tracheae would need to occupy most of his body volume, leaving no room for muscles, organs, or bones. His active muscles — consuming roughly 3–4 litres of oxygen per minute at race pace —

Kipchoge's body during the race. – **Use data or quantitative reasoning where possible (e.g., oxygen demand, heart rate, body size).** – **Conclude with a statement about what this thought experiment reveals about the relationship between structure and function in biology.**	would receive essentially zero oxygen through such a system. Cellular respiration would fail, ATP production would stop, and muscle function would cease. Difference 3 — Tubular Tube-Heart vs. Four-Chambered Heart: The grasshopper's heart is a simple pulsating tube that generates very low pressure and has no separation of circuits. Even if we allowed haemolymph to carry some oxygen (which it does not, in a grasshopper), this pump could not generate the cardiac output Kipchoge needs. At marathon pace, a trained human heart pumps approximately 20–25 litres of blood per minute. A grasshopper's tube-heart, scaled up, could not come close to this. Additionally, with no separation of oxygenated and deoxygenated blood, every 'beat' would mix used and fresh blood, drastically reducing the effective oxygen concentration delivered to tissues. Conclusion — Structure Determines Function: This thought experiment makes one of biology's most important principles vivid: structure and function are inseparable. Every feature of Kipchoge's transport system — the four chambers, the thick left ventricular wall, the high-pressure closed circuit, the valves — exists because it solves a specific problem posed by being a large, warm-blooded, highly active mammal. The grasshopper's system is not inferior — it is perfectly matched to the grasshopper's size, lifestyle, and metabolic needs. When we imagine transplanting one system into the other animal, it fails precisely because form and function co-evolved together. Biology does not produce universal designs; it produces solutions that fit specific organisms in specific environments.

Section 5: Revisiting the Driving Question — Your Conclusion	
Write a concluding paragraph (or short set of paragraphs) that directly and fully answers the driving question: 'How does the structure of an animal's transport system determine its ability to survive and thrive?' Your conclusion should: – **Summarise the main pattern you observed across grasshopper → tilapia → frog → mammal.** – **Explain what 'fit for purpose' means in the context of transport systems.** – **Reflect on what Kipchoge's extraordinary cardiovascular performance reveals about human biology.** – **End with one remaining question you still have, or one new question this unit**	Across the 12 lessons of this unit, one pattern emerged again and again: an animal's transport system is never random — it is precisely matched to the animal's size, metabolic rate, activity level, and environment. Moving from the grasshopper to the tilapia to the frog to Eliud Kipchoge, we see a progression from open to closed circulation, from two chambers to four, from single to double circulation, and from low-pressure to high-pressure delivery. Each step in this progression corresponds to a greater demand for rapid, reliable oxygen and nutrient delivery across a larger, more active body. 'Fit for purpose' means that each system works not because it is the most advanced or the most complex, but because it matches what that particular animal needs to survive in its environment. A grasshopper does not need a four-chambered heart — possessing one would be wasteful and offer no advantage. A marathon runner like Kipchoge, however, could not exist without one. Kipchoge's cardiovascular system is a remarkable example of what the human body can achieve through both evolution and deliberate training. His resting heart rate of 37 bpm, his rapid post-race recovery, and his ability to sustain world-record speeds all point to a heart that has been shaped — both over millions of years of human evolution and over decades of personal training — to be an extraordinarily efficient pump. His left ventricle is thicker and stronger; his stroke volume is larger; his autonomic nervous system responds faster. He is not superhuman — he is a human whose training has pushed his biology to its limits. One question I still have after this unit: If Kipchoge's heart is already so efficient, is there a biological ceiling — a maximum cardiac output — that even the most trained human heart cannot exceed? And if so, what structural feature is the limiting factor?

has sparked for you.	

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Scientific Accuracy of Transport System Concepts	All descriptions of open/closed circulation, heart chamber numbers, double/single circulation, tracheal systems, and cardiac output are accurate and use correct terminology throughout. No factual errors are present.	Most descriptions are accurate and terminology is mostly correct. Minor errors are present but do not undermine the overall explanation.	Several factual errors or misconceptions are present (e.g., confusing open and closed circulation, incorrect chamber counts). Terminology is missing or frequently misused.
Use of Evidence and Data to Support Claims	Claims are consistently supported with specific evidence — including data from the anchoring phenomenon (e.g., Kipchoge's heart rate values, recovery curves), observations from lessons, and comparisons across at least three animal types. Quantitative reasoning is used where appropriate.	Most claims are supported with evidence from lessons or the phenomenon. Some comparisons across animals are included. Quantitative details are occasionally referenced but not consistently.	Claims are mostly unsupported or rely only on general statements. Little or no reference is made to the anchoring phenomenon data or specific lesson observations.
Depth and Completeness of the Grasshopper–Kipchoge Thought Experiment (Section 4)	At least three specific structural differences are identified and each is linked to a precise, scientifically explained consequence for Kipchoge's body during a marathon. Reasoning is logical, detailed, and uses biological mechanisms (e.g., oxygen delivery failure, pressure loss, ATP depletion).	Two or three structural differences are identified with generally correct explanations of consequences. Some mechanistic reasoning is present but lacks full depth or precision in one or two areas.	Fewer than two differences are identified, or consequences are described only vaguely (e.g., 'he would not get enough oxygen') without explaining the biological mechanism. The thought experiment is incomplete.
Connection to the Anchoring Phenomenon and Driving Question	The response consistently refers back to Kipchoge's marathon performance, the heart-rate recovery data, and the animals from the anchoring phenomenon (grasshopper, tilapia, frog). The driving question is answered directly and fully in the conclusion.	The anchoring phenomenon is referenced in most sections. The driving question is addressed in the conclusion but the link to the phenomenon could be stronger or more specific in places.	The anchoring phenomenon is mentioned only briefly or in one section. The driving question is not clearly or fully answered in the conclusion.
Organisation, Clarity, and Scientific Communication	The response is clearly organised across all five sections, written in full sentences, and is easy to follow. Scientific vocabulary is used accurately and appropriately. The explanation flows logically from one section to the next, building a coherent cumulative argument.	The response is mostly organised and readable. Scientific vocabulary is used but occasionally imprecisely. Most sections flow logically, though transitions between ideas could be clearer in places.	The response is difficult to follow, with ideas presented in a disorganised way or as disconnected bullet points without explanation. Scientific vocabulary is largely absent or used incorrectly.